THE UNIVERSITY OF ALABAMA IN HUNTSVILLE

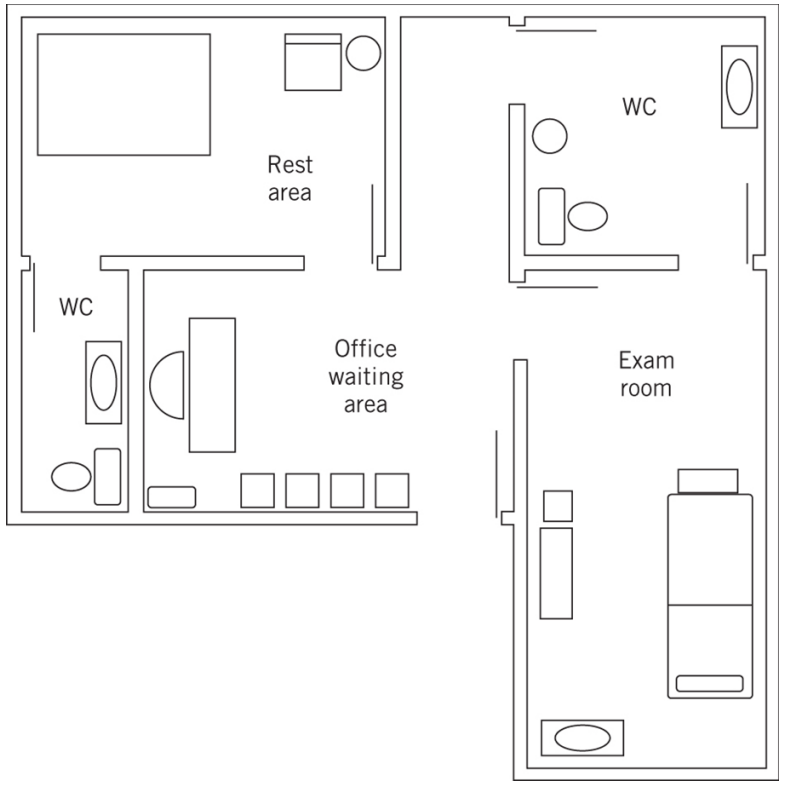
ISE 430/530: Manufacturing Systems and Facilities Design

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Industrial & Systems and Engineering Management Department

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Health Services

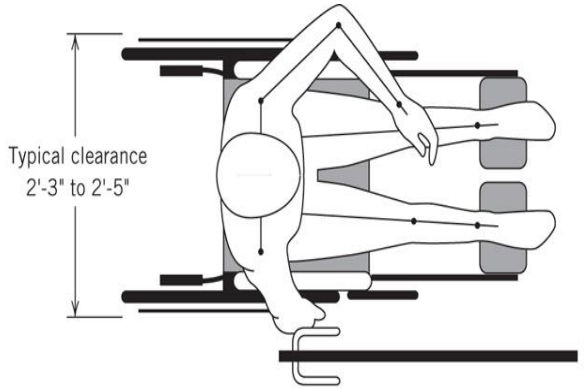
- Health Services
  - Examination room, beds, waiting rooms
  - First aid room
- Service Range
  - Pre-employment examinations, first aid, major medical, dental care, illness treatment
- Minimum requirement - First aid room
  - Bed, approved first aid kit, two chairs (100 ft2)
- Locate close to hazardous tasks but in a quiet and soundproof area



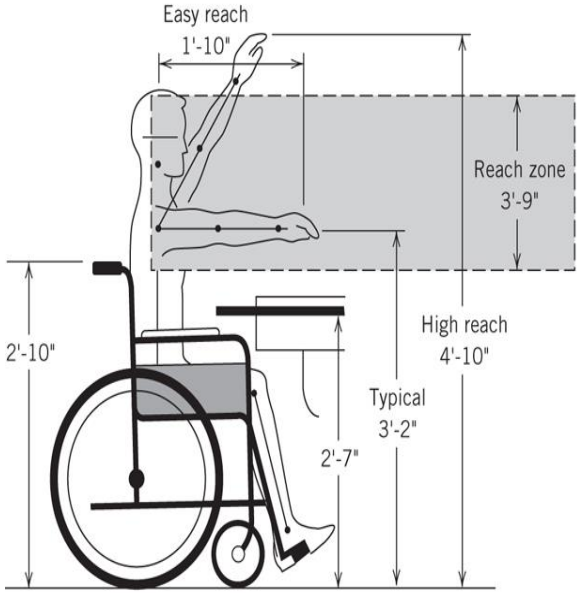
Barrier-Free Compliance

- Americans with Disabilities Act (ADA) requires that disabled persons have same ability to use facility as able bodied individuals
  - Barrier Free Facility – remove all barriers that impede use of facility by disabled persons

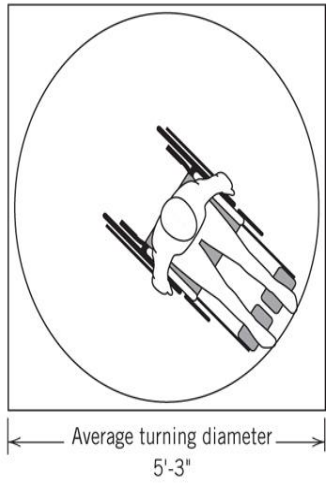
- Barriers
  - Physical object that impedes a disabled person's access to facility


Typical clearance
2'-3" to 2'-5"

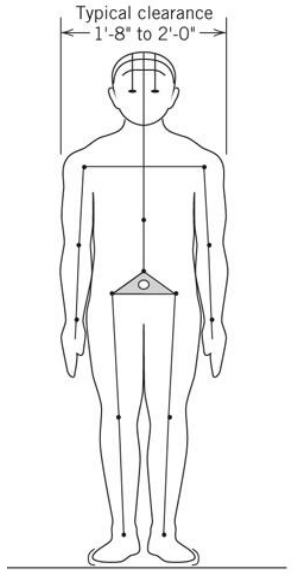
(a)


Easy reach
1'-10"
Reach zone
3'-9"
2'-10"
High reach
4'-10"
Typical
3'-2"
2'-7"

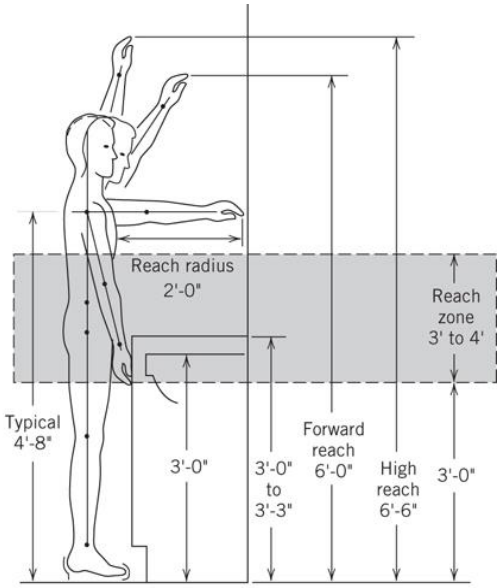
(b)


Average turning diameter
5'-3"

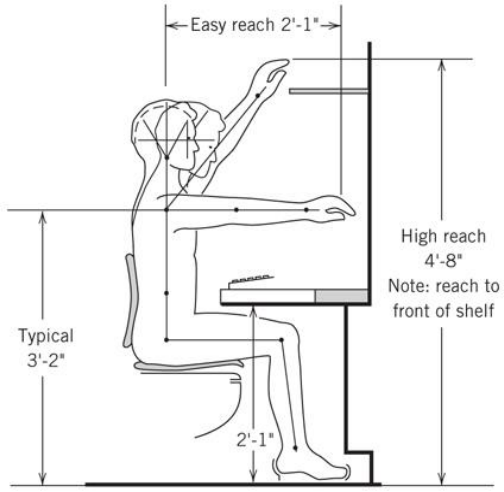
(c)


Typical clearance
1'-8" to 2'-0"

(a)


Reach radius
2'-0"
Reach
zone
3' to 4'
Typical
4'-8"
3'-0"
3'-0"
to
3'-3"
Forward
reach
6'-0"
High
reach
6'-6"
3'-0"

(b)


Easy reach 2'-1"
High reach
4'-8"
Note: reach to
front of shelf
Typical
3'-2"
2'-1"

(c)

RULA scores



RULA Employee Assessment Worksheet

TuMeke Ergonomics

A. Arm and Wrist Analysis

Step 1 : Locate Upper Arm Position



Step 1a: Adjust...
If shoulder is raised: +1
If upper arm is abducted : +1
If arm is supported or person is leaning : -1

Upper Arm Score

Step 2 : Locate Lower Arm Position



Lower Arm Score

Step 2a : Adjust...
If either arm is working across midline or out to side of body: +1

Step 3 : Locate Wrist Position



Step 3a : Adjust...
If wrist is bent from midline : +1

Wrist Score

Step 4 : Wrist Twist
If wrist is twisted in mid-range : +1
If wrist is at or near end of range: +2

Wrist Twist Score

Step 5 : Look-up Posture Score in Table A
Using values from steps 1-4 above, locate score in Table A

Posture Score A +

Step 6 : Add Muscle Use Score
If posture mainly static (i.e. held > 1 minute),
Or if actions repeated occurs 4x per minute : +1

Muscle Use Score +

Step 7 : Add Force/Load Score
If load < 4.4 lbs (intermittent) : +0
If load 4.4 to 22 lbs (intermittent) : +1
If load 4.4 to 22 lbs (static or repeated) : +2
If more than 22 lbs or repeated or shocks : +3

Force/Load Score =

Step 8: Find Row in Table C

Wrist & Arm Score

SCORES

Table A		Wrist Posture Score							
		1		2		3		4	
Upper Arm	Lower Arm	Wrist Twist		Wrist Twist		Wrist Twist		Wrist Twist	
		1	2	1	2	1	2	1	2
1	1	1	2	2	2	2	3	3	3
	2	2	2	2	2	3	3	3	3
	3	2	3	3	3	3	3	4	4
2	1	2	3	3	3	3	4	4	4
	2	3	3	3	3	3	4	4	4
	3	3	4	4	4	4	4	5	5
3	1	3	3	4	4	4	4	5	5
	2	3	4	4	4	4	4	5	5
	3	4	4	4	4	4	5	5	5
4	1	4	4	4	4	4	5	5	5
	2	4	4	4	4	4	5	5	5
	3	4	4	4	5	5	5	6	6
5	1	5	5	5	5	5	6	6	7
	2	5	6	6	6	6	7	7	7
	3	6	6	6	7	7	7	7	8
6	1	7	7	7	7	7	8	8	9
	2	8	8	8	8	8	9	9	9
	3	9	9	9	9	9	9	9	9

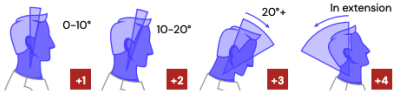
Table C		Neck, trunk and leg score						
		1	2	3	4	5	6	7+
Wrist and Arm Score	1	1	2	3	3	4	5	5
	2	2	2	3	4	4	5	5
	3	3	3	3	4	4	5	6
	4	3	3	3	4	5	6	6
	5	4	4	4	5	6	7	7
	6	4	4	5	6	6	7	7
	7	5	5	6	6	7	7	7
	8+	5	5	6	7	7	7	7

RULA Scoring (Final score from Table C)	
1 or 2	acceptable posture
3 or 4	further investigation, change may be needed
5 or 6	further investigation, change soon
7	investigate and implement change

RULA Score

B. Neck, Trunk and Leg Analysis

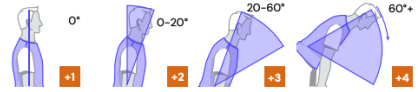
Step 9 : Locate Neck Position



Step 9a: Adjust...
If neck is twisted: +1
If neck is side bending: +1

Neck Score

Step 10 : Locate Trunk Position



Step 10a : Adjust...
If trunk is twisted: +1
If trunk is side bending: +1

Trunk Score

Step 11 : Legs
If legs and feet are supported: +1
If not: +2

Leg Score

Table B		Trunk Posture Score											
		1		2		3		4		5		6	
Neck Posture Score		Legs		Legs		Legs		Legs		Legs		Legs	
		1	2	1	2	1	2	1	2	1	2	1	2
1		1	3	2	3	3	4	5	5	6	6	7	7
2		2	3	2	3	4	5	5	5	6	7	7	7
3		3	3	3	4	4	5	5	6	6	7	7	7
4		5	5	5	6	6	7	7	7	7	7	8	8
5		7	7	7	7	7	8	8	8	8	8	8	8
6		8	8	8	8	8	8	8	9	9	9	9	9

Step 12 : Look-up Posture Score in Table B
Using values from steps 9-11 above, locate score in Table B

Posture Score B +

Step 13 : Add Muscle Use Score
If posture mainly static (i.e.hold > 1 minute),
Or if actions repeated occurs 4x per minute: +1

Muscle Use Score +

Step 14 : Add Force/Load Score
If load < 4.4 lbs (intermittent) : +0
If load 4.4 to 22 lbs (intermittent) : +1
If load 4.4 to 22 lbs (static or repeated) : +2
If more than 22 lbs or repeated or shocks : +3

Force/Load Score =

Step 15 : Find Column in Table C
Add values from steps 12-14 to obtain
Neck, Trunk and Leg Score. Find Column in Table C

Neck, Trunk & Leg Score

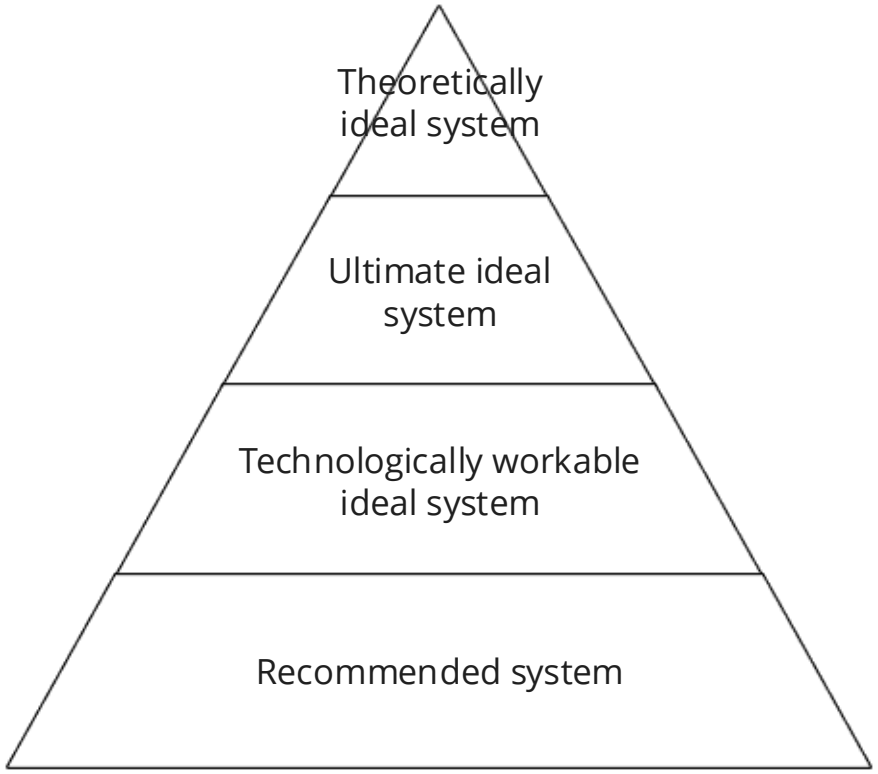
Automate this asssessment with Computer Vision! Find us at www.tumeke.io

based on RULA: a survey method for the investigation of work-related upper limb disorders, McAtamney & Corlett, Applied Ergonomics 1993, 24(2), 91-99

Layout Procedures

Nadler's Ideal Systems Approach

1. ***Aim*** for "theoretically ideal system"
2. ***Conceptualize*** "ultimate ideal system"
3. ***Design*** "technologically workable ideal system"
4. ***Install*** "recommended system"



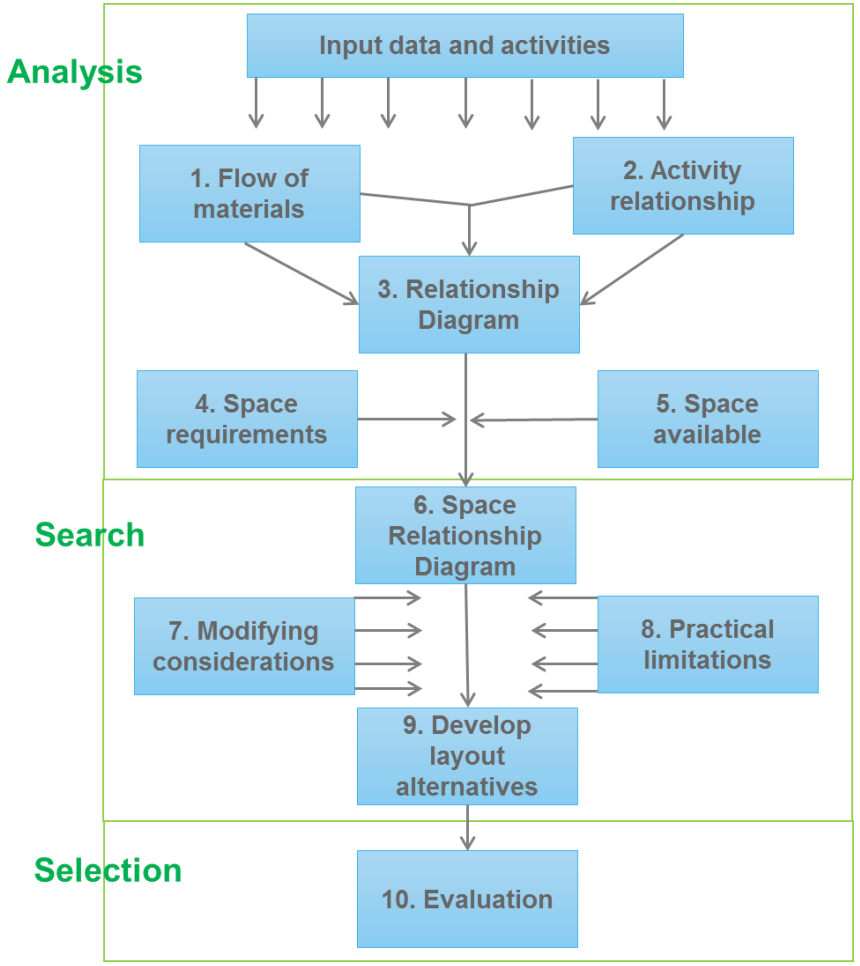
Layout Procedures

- **Immer's Basic Steps**
  1. Put problem on paper
  2. Show lines of flow
  3. Convert flow lines to machine lines

- **Reed's Plant Layout Procedure**
  1. Analyze product or process to be produced
  2. Determine process required to manufacture product
  3. Prepare layout planning chart
  4. Determine work stations
  5. Analyze storage area requirement
  6. Establish minimum isle widths
  7. Establish office requirements
  8. Consider personnel facilities and services
  9. Survey plant services
  10. Provide for future expansion

Layout Procedures

- Muther's systematic layout planning (SLP) procedure



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SLP-block Layout

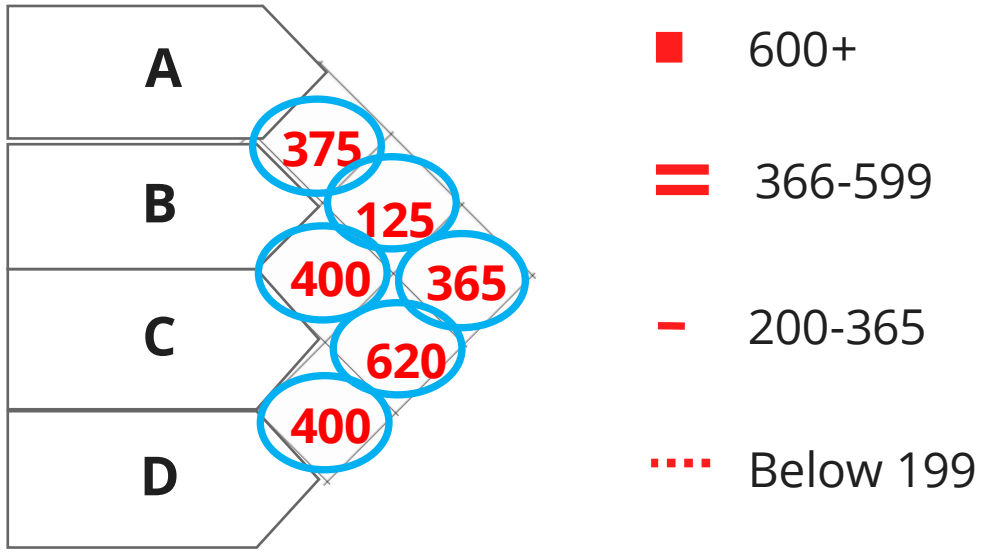
- Exercise: Four departments are to be located in a building of 1000x600 ft. The expected personnel flows and requirement for departments are shown in the table below. Develop a block layout using SLP. (hint: intuitive execution by hand works fine)

Dept.	A	B	C	D
A (200x200 ft)	0	250	25	240
B (400x400 ft)	125	0	400	335
C (600x600 ft)	100	0	0	225
D (200x200 ft)	125	285	175	0

SLP-block Layout

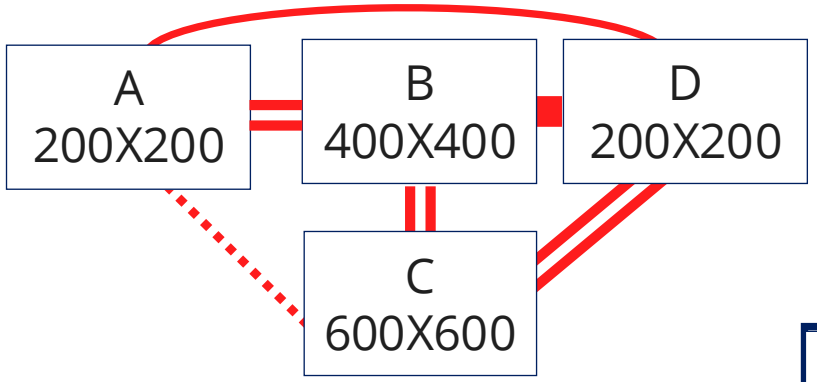
- Developing Activity Relationship Chart

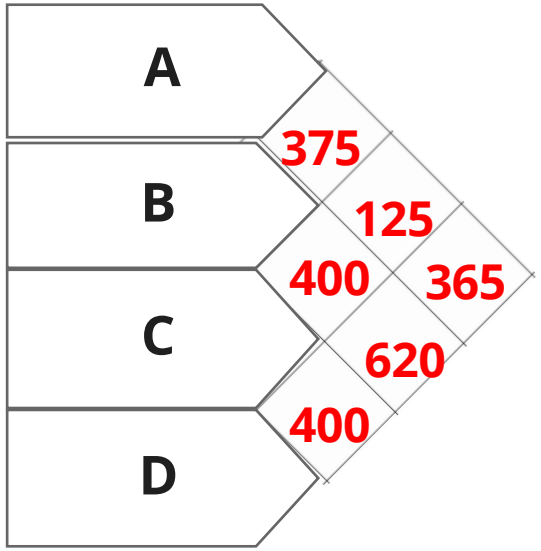
	A	B	C	D
A	0	250	25	240
B	125	0	400	335
C	100	0	0	225
D	125	285	175	0



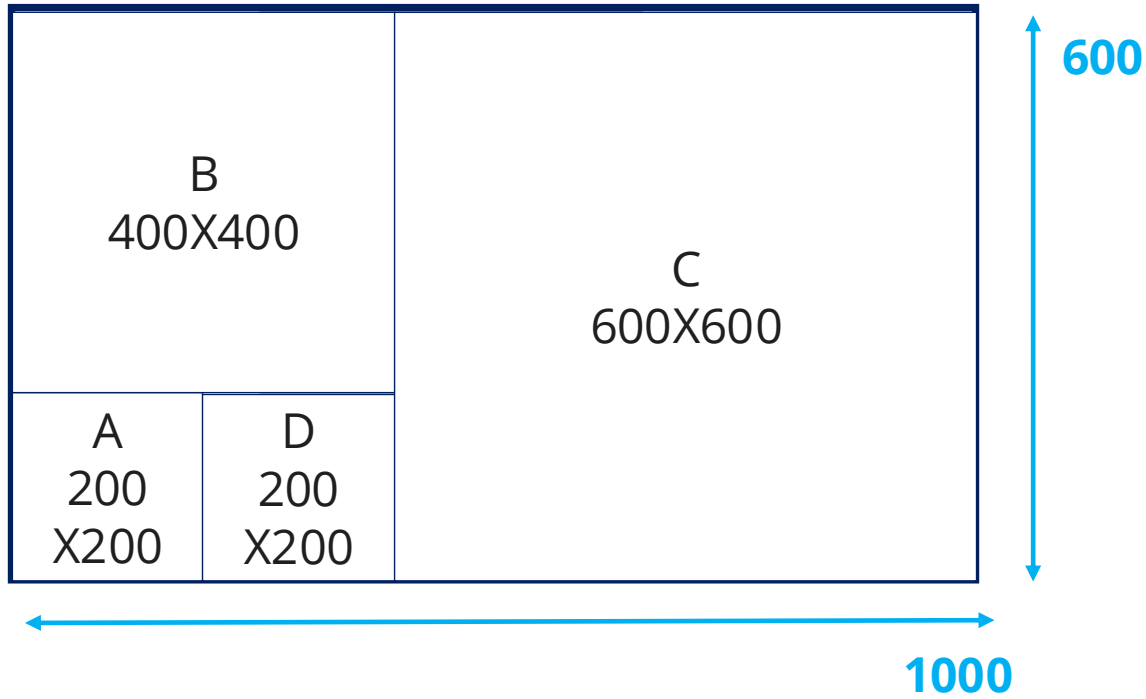
SLP-block Layout

Developing Space Relationship Chart





Dept.
A (200x200 ft)
B (400x400 ft)
C (600x600 ft)
D (200x200 ft)



Classification of Algorithms

Objective Functions 1: Minimize total flow

Called a “distance-based” objective

Distance is measured rectilinearly from the centroid

$$Minimize\ Z = \sum_{i=1}^{m}\sum_{j=1}^{m} f_{ij}\, c_{ij} d_{ij}$$

There are total *m* departments

Flow amount from department i to j

Cost of transporting unit of flow for unit distance from department i to j

distance from department i to j

Classification of Algorithms

Objective Functions 2: Maximize adjacency score

- Called a “adjacency-based” objective
- Sum of all adjacent relationship values
- Easy to use and intuitive
- But disregards distance between non-adjacent departments
- This means you can generate layouts with similar scores but vastly different travel distances

$$Maximize\ Z = \sum_{i=1}^{m} \sum_{j=1}^{m} f_{ij}\, x_{ij}$$

There are total *m* departments

Flow amount from department i to j

Adjacency indicator

$$= \begin{cases} x_{ij} = 1 & \text{if department i is adjacent to department j} \\ x_{ij} = 0 & otherwise \end{cases}$$

Classification of Algorithms

Objective Functions 2: Maximize adjacency score

Normalized Objective Function

$$Maximize\ Z = \frac{\sum_{i=1}^{m}\sum_{j=1}^{m} f_{ij}x_{ij}}{\sum_{i=1}^{m}\sum_{j=1}^{m} f_{ij}}$$

Total possible value: when every department is next to every other department

Incorporating negative value of adjacency

Positive adjacency

Negative adjacency

$$Maximize\ Z = \frac{\sum_{(i,j)\in S=1}^{m} f_{ij}\,x_{ij} + \sum_{(i,j)\in \bar{S}=1}^{m} f_{ij}\,(1 - x_{ij})}{\sum_{(i,j)\in S=1}^{m} f_{ij} - \sum_{(i,j)\in \bar{S}=1}^{m} f_{ij}}$$

Positive department pairs

Negative department pairs

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Classification of Algorithms

Objective Functions 1: Minimize total flow

Flow amount from department i to j

Cost of transporting unit of flow for unit distance from department i to j

distance from department i to j

$$Minimize\ Z = \sum_{i=1}^{m}\sum_{j=1}^{m} f_{ij}\, c_{ij} d_{ij}$$

Flow	A	B	C	D
A	0	250	25	240
B	125	0	400	335
C	100	0	0	225
D	125	285	175	0

Distance

L1 d_{ij}	**A**	**B**	**C**	**D**
A	0	100	150	50
B	100	0	20	15
C	150	20	0	35
D	50	15	35	0

L2 d_{ij}	**A**	**B**	**C**	**D**
A	0	20	35	15
B	20	0	100	50
C	35	100	0	150
D	15	50	150	0

Understanding Objective Functions

Objective Functions 1: Minimize total flow

- $Minimize\ Z = \sum_{i=1}^{m} \sum_{j=1}^{m} f_{ij}\, c_{ij} d_{ij}$

$= f_{12}c_{12}d_{12} + f_{13}c_{13}d_{13} + f_{14}c_{14}d_{14}$

$+ f_{21}c_{21}d_{21} + f_{23}c_{23}d_{23} + f_{24}c_{24}d_{24}$

$+ f_{31}c_{31}d_{31} + f_{32}c_{32}d_{32} + f_{34}c_{34}d_{34}$

$+ f_{41}c_{41}d_{41} + f_{42}c_{42}d_{42} + f_{43}c_{43}d_{43}$

c_{ij} = \$1/unit*foot

100*250 + 150*25 + 50*240
+ 100*125 + 20*400 + 15*335
+ 150*100 + 20*0 + 35*225
+ 50*125 + 15*285 + 175*35
=\$105800

L1 d_{ij}	A	B	C	D
A	0	100	150	50
B	100	0	20	15
C	150	20	0	35
D	50	15	35	0

Flow	A	B	C	D
A	0	250	25	240
B	125	0	400	335
C	100	0	0	225
D	125	285	175	0

Understanding Objective Functions

Objective Functions 2: Maximize adjacency score

- $Maximize\ Z = \sum_{i=1}^{m} \sum_{j=1}^{m} f_{ij}\, x_{ij}$

Flow amount from department i to j

Adjacency indicator

$$= \begin{cases} x_{ij} = 1 & \text{if dptmt i is adjacen to dptmt j} \\ x_{ij} = 0 & otherwise \end{cases}$$

Flow	A	B	C	D
A	0	250	25	240
B	125	0	400	335
C	100	0	0	225
D	125	285	175	0

A-B-C-D

L1 x_{ij}	A	B	C	D
A		adj		
B	adj		adj	
C		adj		adj
D			adj	

A-C-B-D

L2x_{ij}	A	B	C	D
A			adj	
B			adj	adj
C	adj	adj		
D		adj		

Understanding Objective Functions

Objective Functions 2: Maximize adjacency score

- $Maximize\ Z = \sum_{i=1}^{m} \sum_{j=1}^{m} f_{ij}\, x_{ij}$

$$= f_{12}x_{12} + f_{13}x_{13} + f_{14}x_{14}$$
$$+ f_{21}x_{21} + f_{23}x_{23} + f_{24}x_{24}$$
$$+ f_{31}x_{31} + f_{32}x_{32} + f_{34}x_{34}$$
$$+ f_{41}x_{41} + f_{42}x_{42} + f_{43}x_{43} \quad \textbf{=1175}$$

Total number of flow that benefited from "adjacency"

$$Maximize\ Z = \frac{\sum_{i=1}^{m} \sum_{j=1}^{m} f_{ij}\, x_{ij}}{\sum_{i=1}^{m} \sum_{j=1}^{m} f_{ij}}$$

$$= \frac{1175}{2285} = 0.51(\%)$$

A-B-C-D

L1 x_{ij}	A	B	C	D
A		adj		
B	adj		adj	
C		adj		adj
D			adj	

Flow	A	B	C	D
A	0	250	25	240
B	125	0	400	335
C	100	0	0	225
D	125	285	175	0

Understanding Objective Functions

Objective Functions 2: Maximize adjacency score

- $Maximize\ Z = \sum_{i=1}^{m} \sum_{j=1}^{m} f_{ij}\, x_{ij}$

$$= f_{12}x_{12} + f_{13}x_{13} + f_{14}x_{14}$$
$$+ f_{21}x_{21} + f_{23}x_{23} + f_{24}x_{24}$$
$$+ f_{31}x_{31} + f_{32}x_{32} + f_{34}x_{34}$$
$$+ f_{41}x_{41} + f_{42}x_{42} + f_{43}x_{43} \quad \textbf{=1145}$$

Total number of flow that benefited from "adjacency"

$$Maximize\ Z = \frac{\sum_{i=1}^{m} \sum_{j=1}^{m} f_{ij}\, x_{ij}}{\sum_{i=1}^{m} \sum_{j=1}^{m} f_{ij}}$$

$$= \frac{1145}{2285} = 0.50(\%)$$

Layout Alternative 1 is a better one.

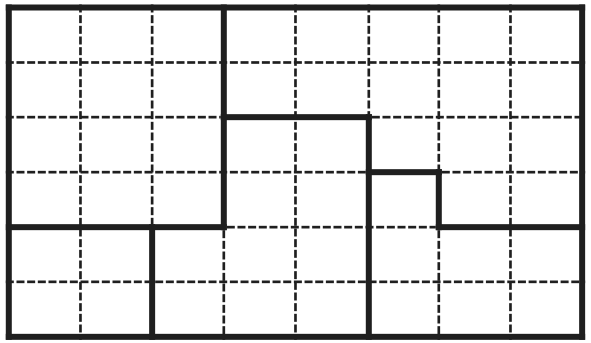
A-C-B-D

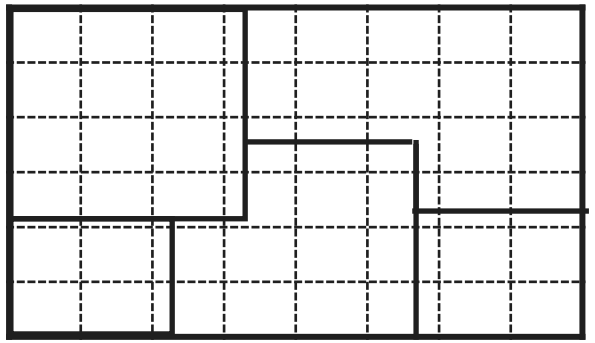
L2x_{ij}	A	B	C	D
A			adj	
B			adj	adj
C	adj	adj		
D		adj		

Flow	A	B	C	D
A	0	250	25	240
B	125	0	400	335
C	100	0	0	225
D	125	285	175	0

Classification of Algorithms

- Layout representation format
  - Discrete
    - The area of each department is rounded off to the nearest integer number of grids.
    - A smaller grid size yields a finer resolution and gives more flexibility in department shapes but results in a larger number of grids which complicates the computations.
  - Continuous
    - Doesn't use a grid
    - More flexible but more difficult to use
    - Usually limited to rectangular building and departments





Classification of Algorithms

- Primary functions
  - Construction-based
    - Develop ‘from scratch' and progressively build layout.
    - Building dimensions are given (moving into a new building) VS flexible building dimension (“green field” application, site plan is given)
  - Improvement-based
    - Start with an initial layout and try to improve it through incremental changes
  - Construction and Improvement-based

Layout Algorithms

- Pairwise exchange method
- Graph-based method
- CRAFT (Computerized Relative Allocation of Facilities Technique)
- BLOCPLAN
- MIP
- LOGIC (Layout Optimization with Guillotine Induced Cuts)
- MULTIPLE (Multifloor Plant Layout Evaluation)

Pairwise Exchange Method

- Algorithm:
  - Compute the total cost of existing layout based on initial from-to chart and distance matrix.
  - Exchange two departments (for all possible combinations given current layout solution)
  - Recompute the total cost
  - Continue until you find the lowest total cost
- Remarks:
  - Iterative procedure
  - Not guaranteed to find optimal solution but will be locally optimal.
  - Selecting the largest reduction in total cost for each iteration is equivalent to “steepest descent” in optimization
  - Why don’t we just check all possible combinations?
  - When we have n department, n! combinations exist.

Pairwise Exchange Method

Iteration 1



	A	B	C	D
A		5	8	10
B			5	3
C				2
D				

From-to chart

	A	B	C	D
A		15	35	50
B			20	35
C				15
D				

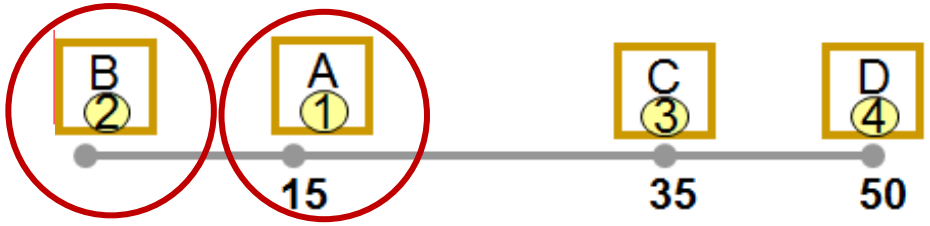
Distance matrix

TC_{ABCD} = 5(15)+8(35)+10(50)+5(20)+3(35)+2(15)=1090 **Initial cost**

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Pairwise Exchange Method

Iteration 1



	A	B	C	D
A		5	8	10
B			5	3
C				2
D				

From-to chart

	A	B	C	D
A		15	20	35
B			35	50
C				15
D				

Distance matrix

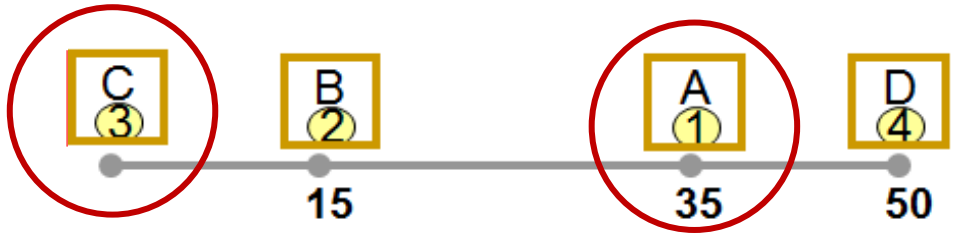
$TC_{ABCD} = 5(15)+8(35)+10(50)+5(20)+3(35)+2(15)=1090$ **Initial cost**

$TC_{BACD} = 5(15)+8(20)+10(35)+5(35)+3(50)+2(15)=940$

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Pairwise Exchange Method

Iteration 1



	A	B	C	D
A		5	8	10
B			5	3
C				2
D				

From-to chart

$TC_{ABCD} = 5(15)+8(35)+10(50)+5(20)+3(35)+2(15)=1090$

$TC_{BACD} = 5(15)+8(20)+10(35)+5(35)+3(50)+2(15)=940$

$TC_{CBAD} = 5(20)+8(35)+10(15)+5(15)+3(35)+2(50)=810$

$TC_{DBCA} = 5(35)+8(35)+10(15)+5(20)+3(35)+2(50)=1010$

$TC_{ACBD} = 5(35)+8(15)+10(50)+5(20)+3(15)+2(35)=1010$

$TC_{ADCB} = 5(50)+8(35)+10(15)+5(15)+3(35)+2(20)=900$

$TC_{ABDC} = 5(15)+8(50)+10(35)+5(35)+3(20)+2(15)=1090$

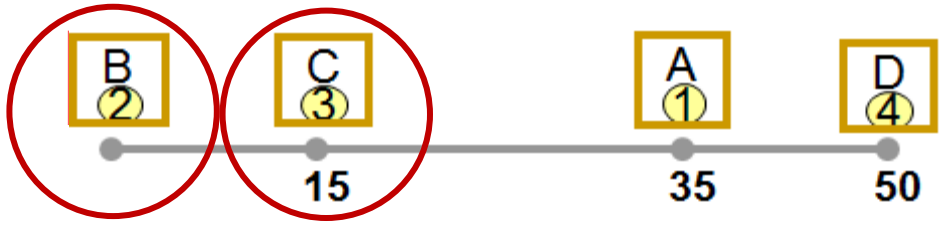
	A	B	C	D
A		20	35	15
B			15	35
C				50
D				

Distance matrix

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Pairwise Exchange Method

Iteration 2



	A	B	C	D
A		5	8	10
B			5	3
C				2
D				

From-to chart

	A	B	C	D
A		35	20	15
B			15	50
C				35
D				

Distance matrix

$TC_{CBAD} = 5(20)+8(35)+10(15)+5(15)+3(35)+2(50)=810$

$TC_{BCAD} = 5(35)+8(20)+10(15)+5(15)+3(50)+2(35)=730$

$TC_{ABCD} = 5(15)+8(35)+10(50)+5(20)+3(35)+2(15)=1090$

$TC_{DBAC} = 5(20)+8(15)+10(35)+5(35)+3(15)+2(50)=890$

$TC_{CABD} = 5(20)+8(15)+10(35)+5(35)+3(15)+2(50)=890$

$TC_{CDAB} = 5(15)+8(35)+10(20)+5(50)+3(35)+2(15)=940$

$TC_{CBDA} = 5(35)+8(50)+10(15)+5(15)+3(20)+2(35)=930$

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Pairwise Exchange Method

Iteration 3



	A	B	C	D
A		5	8	10
B			5	3
C				2
D				

From-to chart

	A	B	C	D
A		35	20	15
B			15	50
C				35
D				

Distance matrix

$TC_{BCAD} = 5(25)+8(20)+10(15)+5(15)+3(50)+2(35)=730$

$TC_{CBAD} = 5(20)+8(35)+10(15)+5(15)+3(35)+2(50)=810$

$TC_{ACBD} = 5(50)+8(15)+10(35)+5(35)+3(20)+2(20)=995$

$TC_{DCAB} = 5(15)+8(20)+10(35)+5(35)+3(50)+2(15)=940$

$TC_{BACD} = 5(15)+8(20)+10(35)+5(35)+3(50)+2(15)=940$

$TC_{BDAC} = 5(35)+8(15)+10(20)+5(50)+3(15)+2(35)=860$

$TC_{BCDA} = 5(50)+8(35)+10(15)+5(15)+3(35)+2(20)=900$

Current cost from i2

Cannot make further improvements: Stop the process and select BCAD as the solution

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Layout Algorithms

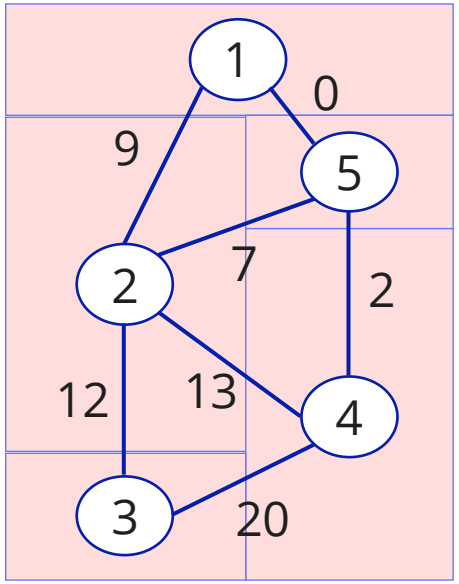
- Pairwise exchange method
- Graph-based method
- CRAFT (Computerized Relative Allocation of Facilities Technique)
- BLOCPLAN
- MIP
- LOGIC (Layout Optimization with Guillotine Induced Cuts)
- MULTIPLE (Multifloor Plant Layout Evaluation)

Graph-based Methods

Construct adjacency graph

Node (vertices): department

Link (arcs, edges): connection between two departments



Represent relationship with weights on links.

Graph-Based Construction Methods

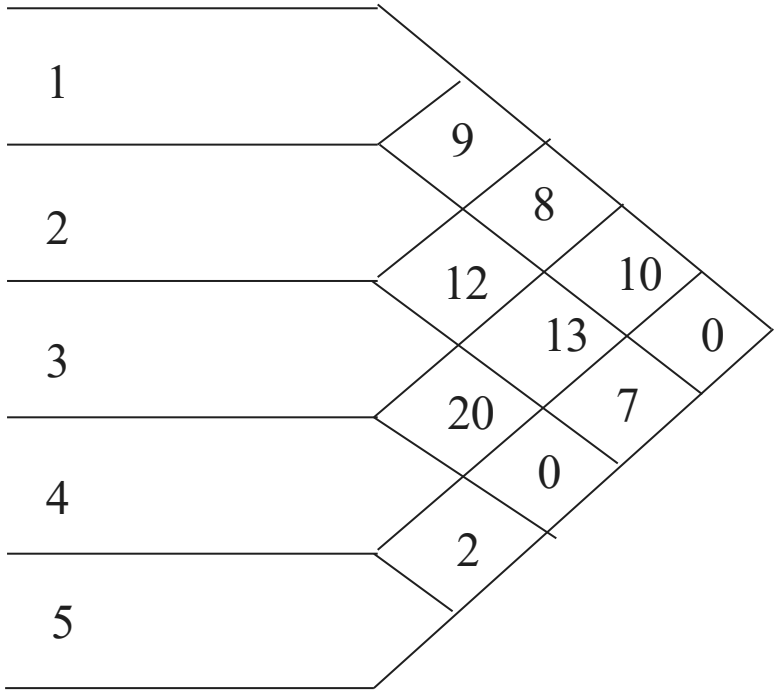
- Approach concerned with adjacency (size of department is not considered here) – maximize adjacency.
- The exterior of the facility counts as a department.
- Each node represents a department.
- A connecting arc between two nodes indicate that two departments share a common border.

No crossing of arcs

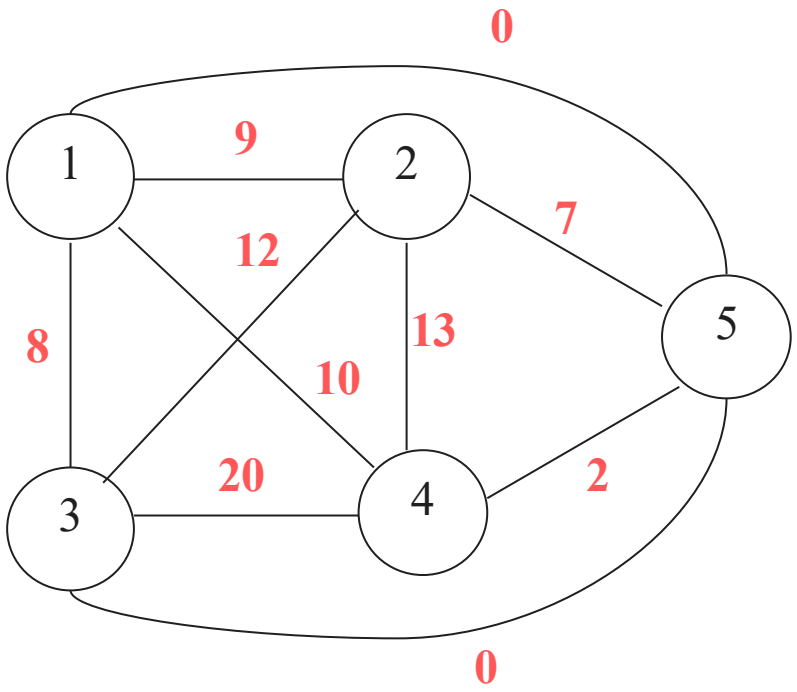
Graph-Based Construction Methods

- Relationship chart with weights is used to assign the weights to each arc.
- The layout with the highest score is chosen. The score is calculated by adding the individual weights of arcs.
- The score only accounts for adjacency relationships, not distance or relationships that are not adjacent.

Graph-Based Construction Methods



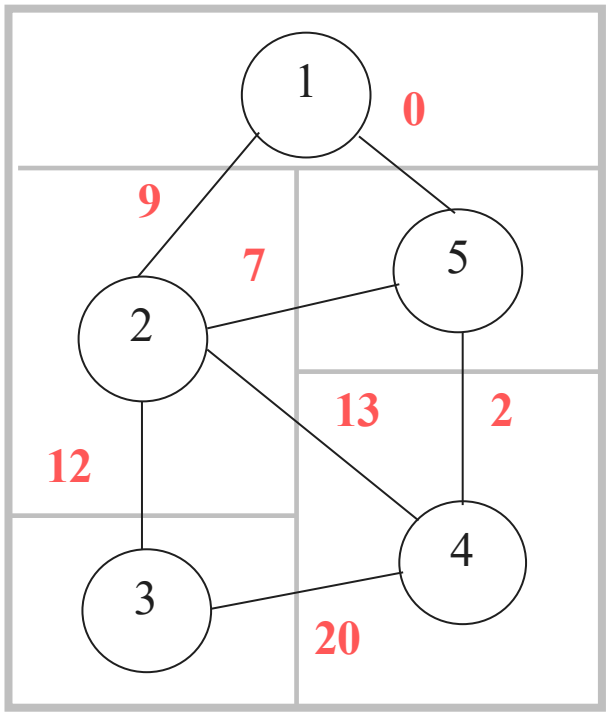
Relationship Chart



Relationship Diagram

Graph when every department is connected to all other departments

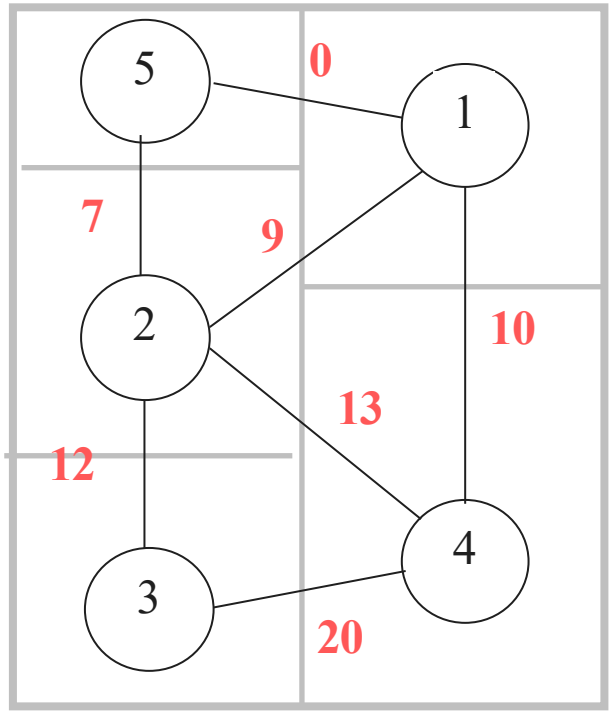
Adjacency Graph for Alternative I



Arc	Weight
1-2	9
1-5	0
2-3	12
2-4	13
2-5	7
3-4	20
4-5	+ 2
	63

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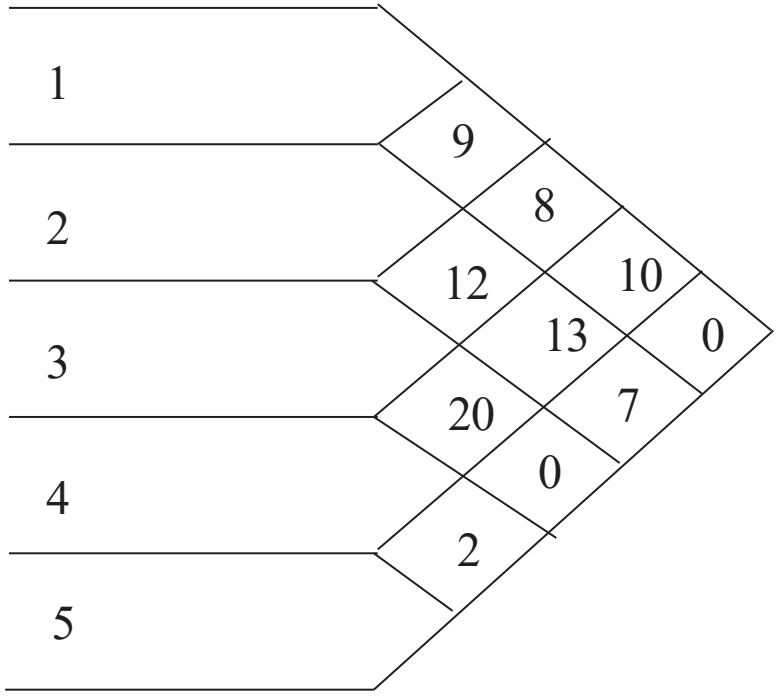
Adjacency Graph for Alternative II

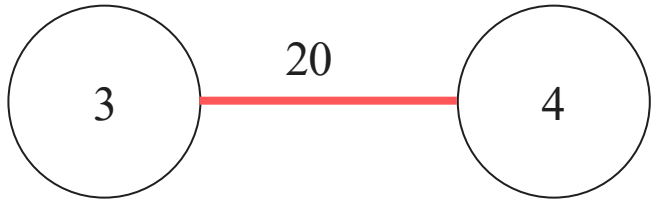


Arc	Weight
1-5	0
1-2	9
1-4	10
2-4	13
2-5	7
2-3	12
3-4	20
	+ 71

Heuristic-Graph Based Layout

•1. From relationship chart, select the department pair with largest weight to begin the graph.

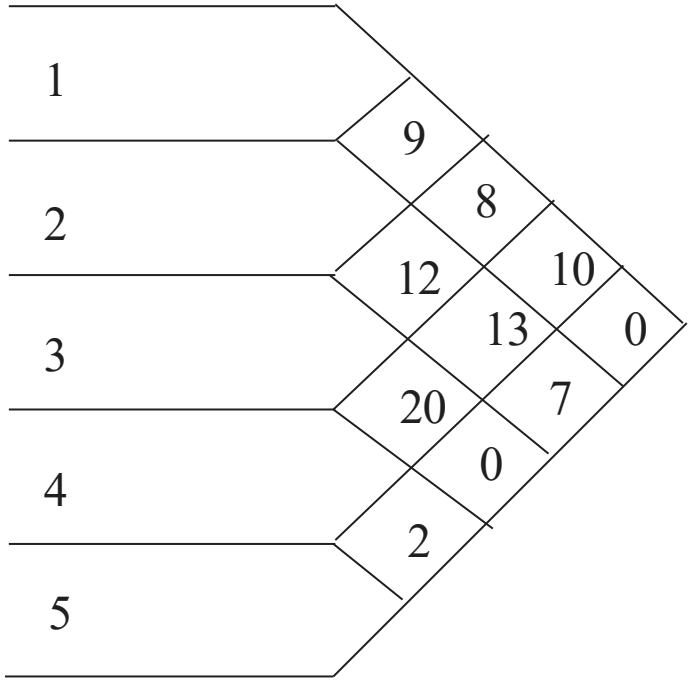




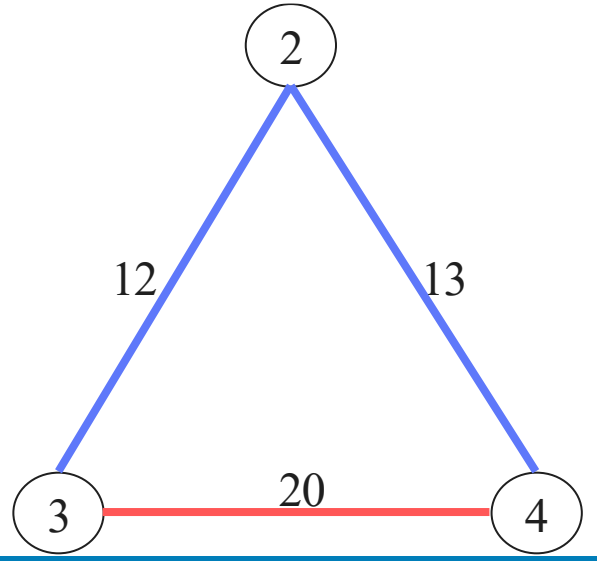
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Heuristic-Graph Based Layout

•2. Select the department with the largest sum of weights with respect to the department pair from previous step.



	3	4	Total
1			
2			
5			

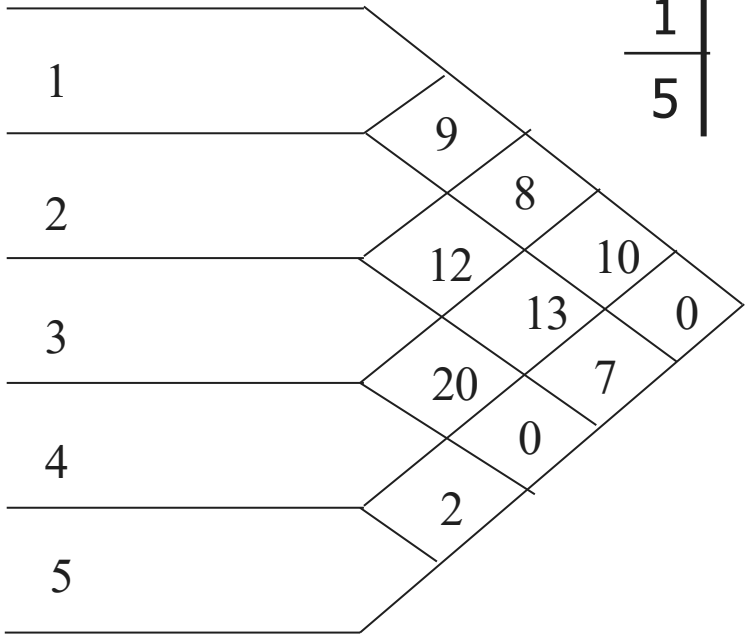


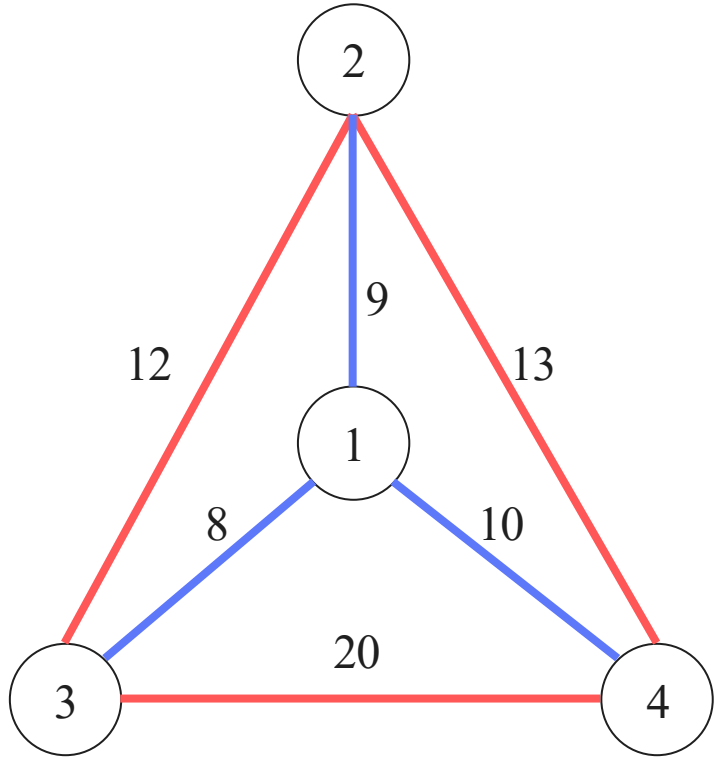
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Heuristic-Graph Based Layout

•3. Select the department with the largest sum of weights with respect to the departments chosen in the previous step.

	2	3	4	Total
1				
5				

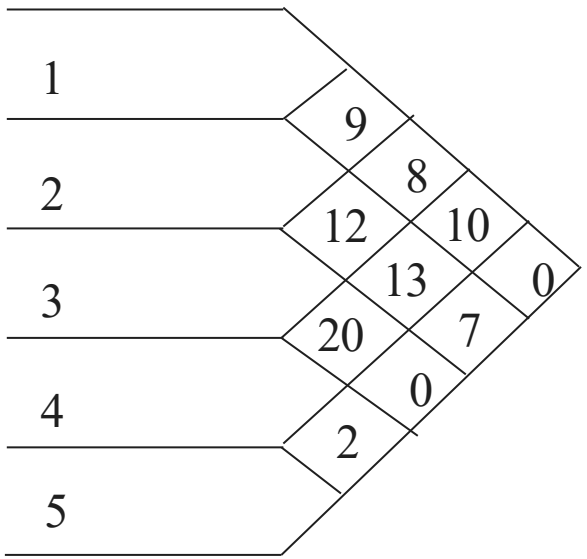




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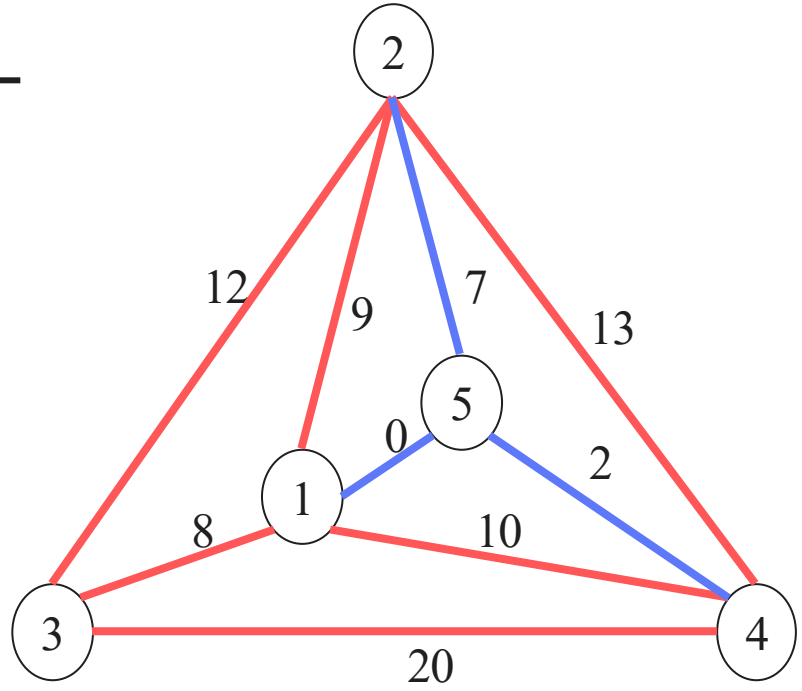
Heuristic-Graph Based Layout

•4. Select the department with the largest sum of weights with respect to the departments chosen in the previous step.



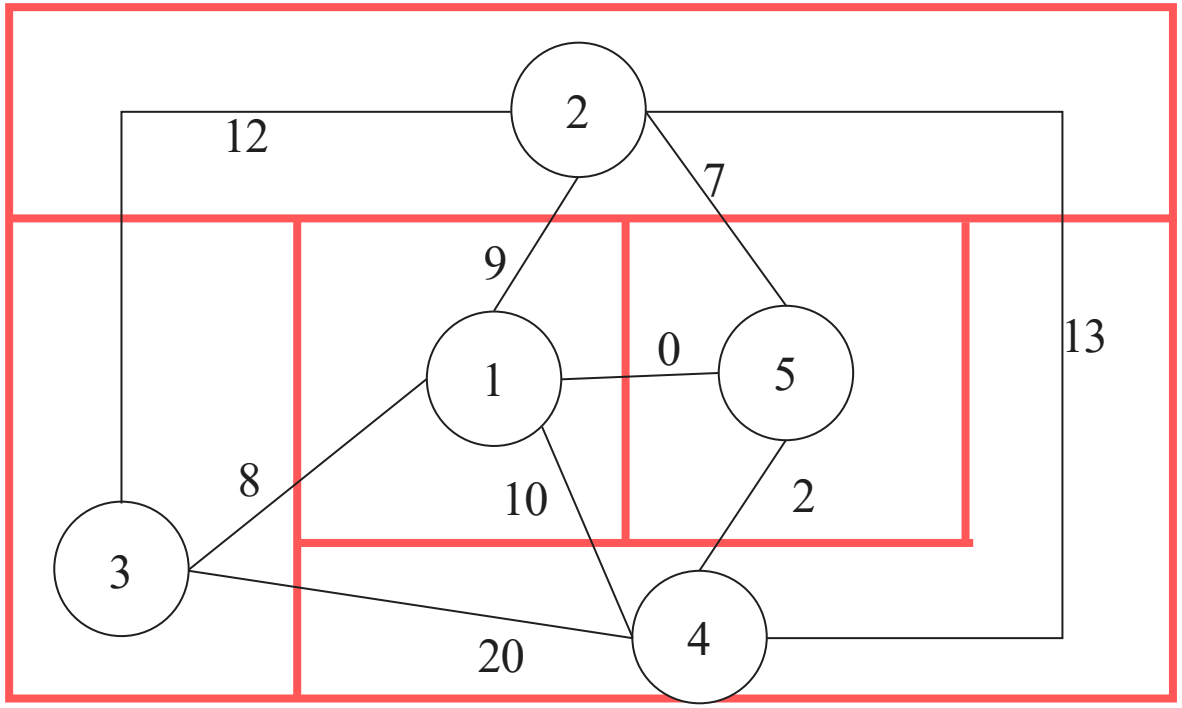
	1	2	3	4
5				

Faces	Total
1-2-3	7
1-2-4	9 (best)
1-3-4	2
2-3-4	9 (best)



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Final Graph Based Layout



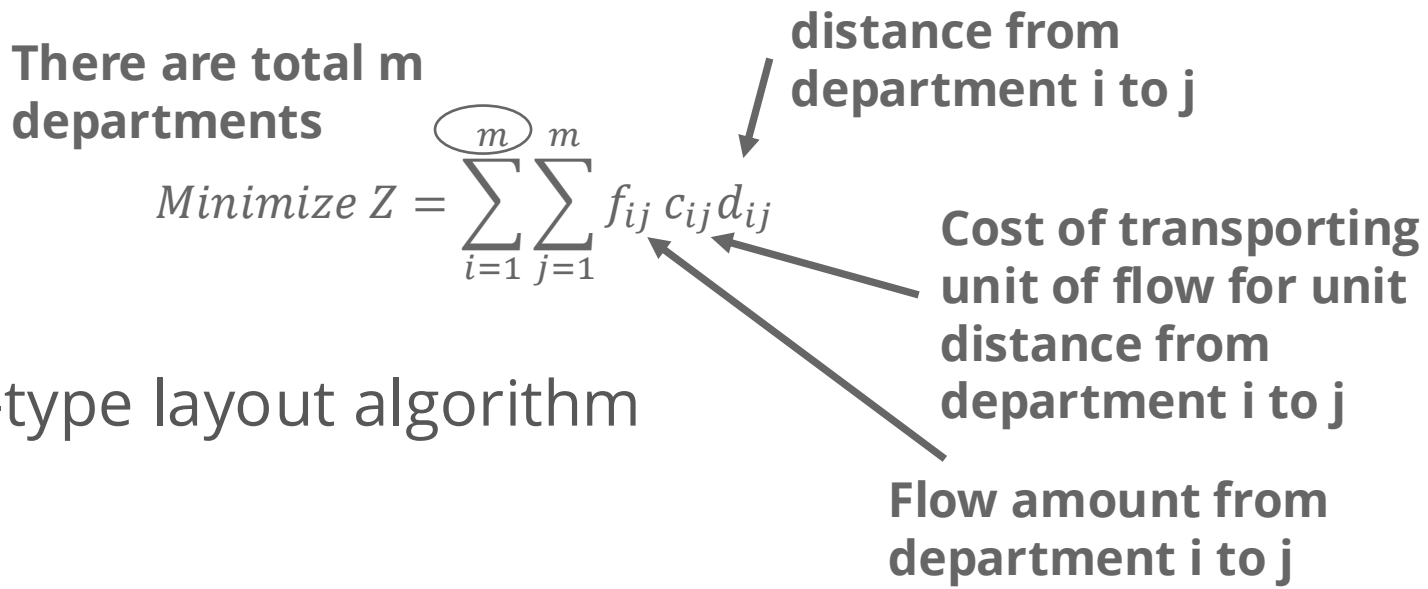
Arc	Weight
1-2	9
1-3	8
1-4	10
1-5	0
2-3	12
2-4	13
2-5	7
3-4	20
4-5	2
Sum	81

Layout Algorithms

- Pairwise exchange method
- Graph-based method
- CRAFT (Computerized Relative Allocation of Facilities Technique)
- BLOCPLAN
- MIP
- LOGIC (Layout Optimization with Guillotine Induced Cuts)
- MULTIPLE (Multifloor Plant Layout Evaluation)

CRAFT

- CRAFT: computerized relative allocation of facilities technique
- First computer-aided layout algorithm (1963)
- The input data is represented in the form of a from-to chart, or qualitative data.
- The main objective behind CRAFT is to minimize total transportation cost:



- Improvement-type layout algorithm

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CRAFT

- CRAFT estimates the new objective function
  - Does not swap departments first and then calculate
  - Results in estimation error if departments are not equal in area
- Adjacency is a necessary but not sufficient criteria for swapping departments
- Quality of final solution depends on the initial layout
- Use dummy departments
  - To fill extra spaces or build irregularities
  - To evaluate aisle location or obstacles/unusable areas (stairways, elevators, etc.)

CRAFT

- CRAFT is a “steepest descent” algorithm
- It doesn’t look ahead or back
- Therefore, the solution may be locally optimal, but not globally optimal
- Because departments can be irregular, the shapes can deteriorate as the number of iterations increase

CRAFT

- Inputs
  - Building dimensions
  - Department dimensions
  - Flow between departments (From-to matrix)
  - Cost of material movement per unit distance (Unit-cost matrix)
  - Initial layout
  - Layout restrictions

Steps in CRAFT

- Calculate centroid of each department and rectilinear distance between pairs of departments centroids (stored in a distance matrix).
- Find the cost of the initial layout by multiplying the from-to chart, unit cost matrix and the distance matrix.
- Improve the layout by performing all-possible two or three-way exchanges.
  - At each iteration, CRAFT selects the interchange that results in the maximum reduction in transportation costs. These interchanges are continued until no further reduction is possible.
- If possible, make adjustments (“Massaging”) for departments layout to be practical.